Tomato Grafting

CROPS SPECIFICS

the MARKET GARDENER

Grafting is a proven, natural strategy that allows us to plant tomatoes in the same place year after year without running into problems with soil-borne diseases. It's a common horticultural technique where two different cultivars are joined together as one plant. The upper part is a cultivar with the desired fruit and is called the scion, while the lower part is called the rootstock, a plant that has been selected because of the vigor of its root system.

Grafting your tomato plants with disease-resistant tomato rootstocks improve yields and overall crop health. If you're planning on growing greenhouse tomatoes seriously, it's a no-brainer (and it's also not difficult!).

MATERIALS NEEDED

→ Rootstock seedlings started in 128 trays (but seeded in half the cells)

→ Grafting seeds started in 128 trays

→ Leakproof trays

→ Disinfectant or 1:10 bleach solution

→ Razor blades

→ Silicone grafting clips of various sizes (1.3 mm [20%], 1.6 mm [60%] and 2.1 mm [20%])

→ Vaporizing spray bottle

→ Plastic domes

→ Humidifier

→ Heater

→ Shop lights with full spectrum fluorescent tubes

THE HEALING CHAMBER

After the graft, the seedlings will need to rest for several days in a high humidity-temperature environment with no light (for the first 2 days) so they don't respire before a vascular connection takes place between the grafted tissues. The humidity rate should be between 90–95% and the ideal temperature between 70–79°F (21–28°C). Any setup that maintains the plants in those conditions will work. We call this room a healing chamber.

This space, or racking * will need artificial lighting and should be very close to where you graft so that the seedlings are displaced as little as possible.

* Plans for a healing shelf unit can be found on the internet.

CHOICE OF ROOTSTOCK

We've used Maxifort with success, but other vigorous, vegetative rootstock should work just as well. Beaufort and Eldorado are other notable ones.

SOWING

Sow seeds 6–8 weeks before your desired transplant date. Grafted tomato plants take 1–2 weeks longer to reach the transplant stage because they stop growing during the healing process—you'll need to factor this into your seeding calendar. Overseed by at least 50% more than the number of plants you plan to transplant. That safety factor is important to counteract any mortality that might occur in the grafting process.

We seed both the rootstock and the different grafted tomatoes (again, called scion) in 128 cell trays. The rootstocks, however, are sowed in one out of two cells of the tray (64 plants/tray). This will increase maneuverability when grafting.

Good germination is ensured by maintaining soil temperature at 80 °F (27 °C). Germination should begin after 3–4 days. We use heating mats with a soil probe to achieve this optimal condition.

Once the seeds have germinated, after about 10 days, we reduce the soil temperature of heating mats to 64–66°F (18–19°C) to encourage a stocky growth habit.

NB. When grafting eggplants, eggplants need to be seeded earlier than their rootstock. We sow the rootstocks for these eggplants once the grafts begin to emerge—approximately 1 week later.

PREPARING THE GRAFT

Plants will be ready for grafting approximately 17 to 21 days after being sown. The best way to know if the plants have reached the right size is to place a grafting clip around the stem. The clip must be well-adjusted. Use the 1.6mm clip for this step, since it is the recommended grafting size for the stem.

At this point, if either the rootstocks or the scions are bigger than the other you can leave them for 24 to 48 hours in a dark place (a closed Rubbermaid bin, a dark room, etc.) so to leave the other to catch up.

Remove seedlings from the greenhouse and place them in a cooler and shadier area close to where they will be grafted. We do this to reduce respiration, halting the growth of the plant for it to put all its energy in the healing that will occur after the graft.

Stop watering the seedlings 24 hours before the graft. This step is crucial to reduce the risks of guttation (water accumulating at the graft line that can lead to a rejection of the scion).

Prepare your operation room. The space should be well lit, but in an area where no direct sunlight will hit the seedlings. Ideally, this should be done indoors or in a covered and temperature-controlled area. Do not graft near a fan or a draft. Any place in a greenhouse will work, as long as it's shaded and doesn't get too hot.

The day of the graft, disinfect your work area with a hand sanitizer. Wash your hands and start with new blades and grafting clips. If you pick up a pathogen, you may transfer it to all of your plants, so overdo it on the cleanliness if needed.

THE GRAFT

On a table, place your 2 trays in front of you with all your material at arm's length.
Double the tray of the rootstock with a leakproof tray so that it can be bottom watered if necessary.

Take a rootstock and with your razor blade, cut it at more than a 45° angle, just above the cotyledons. An angle of 60–70° is ideal since it creates more contact surface. Discard the upper part.

Now choose a scion with a similar diameter and cut it at the same angle under the cotyledons. If the angles aren't compatible, or if the stems are not the same size, you may recut one of the two seedlings for a better fit.

Take a clip of the right size from your boxes, set it on the scion 2⁄3 of the way and place it over the rootstock moving the clip downward to connect both plants. Work slowly making sure that the two parts are firmly touching. Voila ! you've now grafted two plants.

Work from left to right, starting from the middle of your tomato tray and moving towards you. Do one row at a time (from right to left) moving downwards towards you. This way of working minimizes the chances of you disturbing the grafted plants while you're working.

As soon as a tray is full, vaporize the seedlings and cover them with a plastic dome. Generously spray the interior of the dome before closing it.

Make sure your tray is well labelled and carefully transfer it to your healing chamber and start over with another tray.

RECOVERY

Recovery will happen in an environment where humidity is close to 100%, with temperatures around 80–82°F (26–28°C) for the first day and 75-79 °F (24-26 °C) for day 2. **For the first 2 days,** the plants need to be kept in total darkness. Under these conditions, the seedlings need to focus completely on their recovery (≠photosynthesis).

On day 3, turn on artificial lights for 2 hours. Make sure that the shop lights are close enough to the seedlings, otherwise, it may cause the grafted plants to lean towards the light, pulling the graft apart. Plan to check seedling 3 times daily. If they are wilting, raise the humidity level by vaporizing inside your domes. Be gentle—a powerful spray can knock the tops of the grafted plants off.

From this point on, the temperature can be lowered to 72-75°F (22–24°C).

The lights can increase the temperature in the healing chamber. The heater needs to be adjusted.

On day 4, turn on artificial lights for 4 hours. Open the dome/shelter to verify that the soil is still moist. Up to this point, the seedlings consume very little water, but if you feel that it's to dry you can "bottom water" your plants by filling your sub-tray with water. Seedlings will suck up the water as they need it.

On day 5, turn on artificial lights for 6 hours. Open the dome/shelter slightly (1" -2.5 cm) to allow some of the humidity to escape. Check on the seedlings frequently. If they wilt, close the domes to keep the humidity in and start over the following day.

On day 6, turn on artificial lights for 8 hours. If the seedlings are doing well, open the domes even more (2" -5 cm) as long as they don't wilt. If so, keep them at the same stage as day 5 and try again the next day. At this point, bring down the humidity levels to around 65–80%. The humidity cannot be upheld indefinitely otherwise the graft won't be able to merge and will end up developing adventitious roots.

Bring temperatures down a notch, aiming for 68–72°F (20–22°C). The idea is to gradually return the grafted seedlings to the same temperature and humidity levels of your greenhouse.

On day 7 or 8, start acclimatizing the seedlings to normal lighting, turning your lights off for the nights.

After having spent one day at normal light, humidity and temperature, the seedlings can do without their domes and are ready to return to the greenhouse. Make sure the transfer is performed early in the morning and ideally on a day that is not too hot. Return to check that everything is going well.

A few days after their return to the greenhouse, the seedlings are ready to potted up in 6” (15cm) pots.

The silicone grafting clips will widen as the seedling grows and will eventually fall off.

CALENDAR SUMMARY

- → Day 1 and 2 : Dome closed and lights off. Temperatures are kept between 80–82°F (26–28°C) and humidity is close to 100%.
- → Day 3 : Dome closed and lights on for 2 hours, temperature can be lowered to 72–75°F (22-24°C).
- → Day 4 : Dome closed and lights on for 4 hours. Test to see if the graft is starting to take. Temperature between 72–75°F (22–24°C).
- → Day 5 : Domes slightly opened 1” (2.5 cm) and lights on for 6 hours. Temperature between 72–75°F (22–24°C).
- → Day 6 : Domes opened wider 2” (5 cm) and lights on for 8 hours. Temperature between 68-72°F (20-22 °C). Humidity is around 65–80%.
- → Day 7 : Return to normal propagation conditions; lights on for 10 hours, off at night. Temperatures around 68–72°F (20–22°C). Domes are now removed.
- → Day 8 : Transfer trays to greenhouse.
- → Days 9 and 10 : Possible potting up.
- → Days 13 to 17 : Pinch off the heads when the graft is healed and the seedlings have started to grow back. Cut on top of the first two true leaves.
- → Day 25 : Space out plants in the nursery in order to have a density of 3 plants per square foot (8 plants/m2).
- → Day 35 : transplant plants in the greenhouse when they have reached 8 to 10 leaves. Do not bury the grafting line.